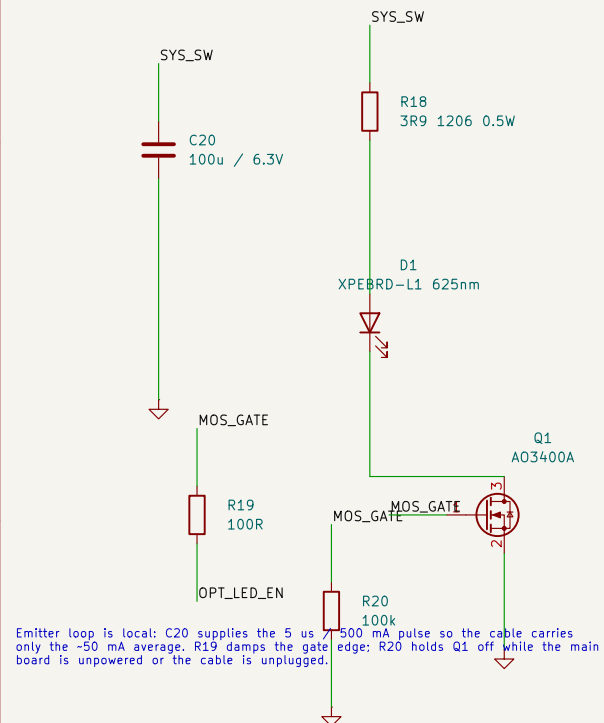
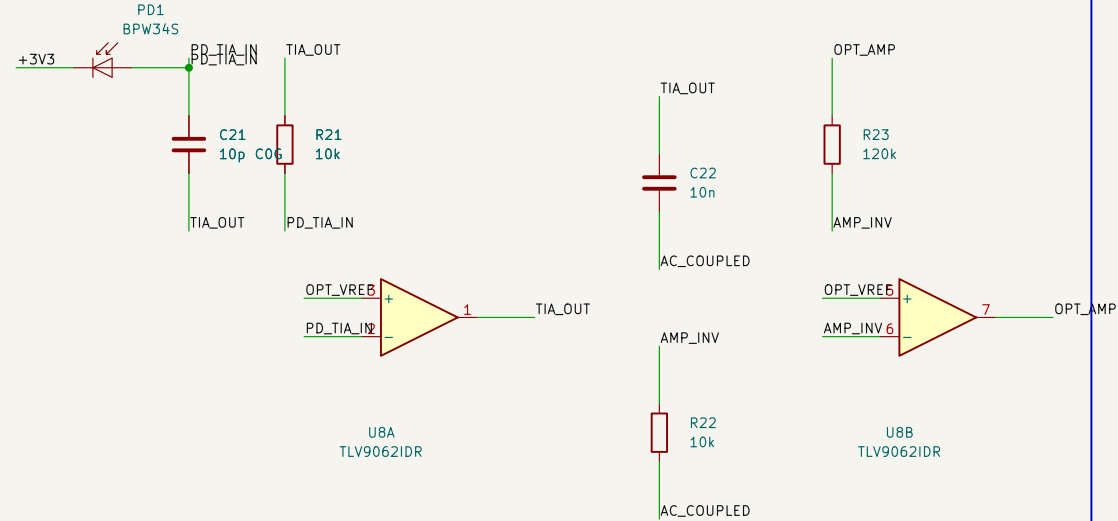


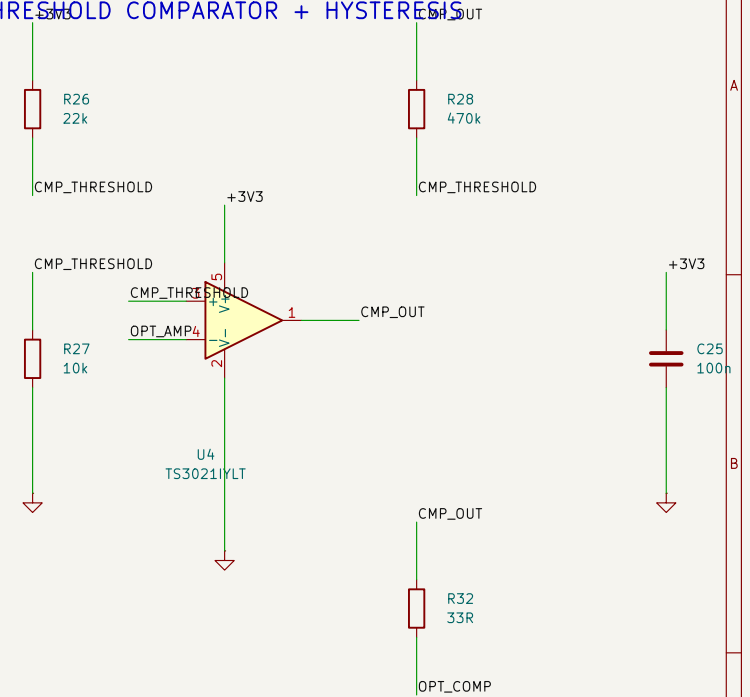
PULSED EMITTER – 20 kHz / 5  $\mu$ s / ~500 mA



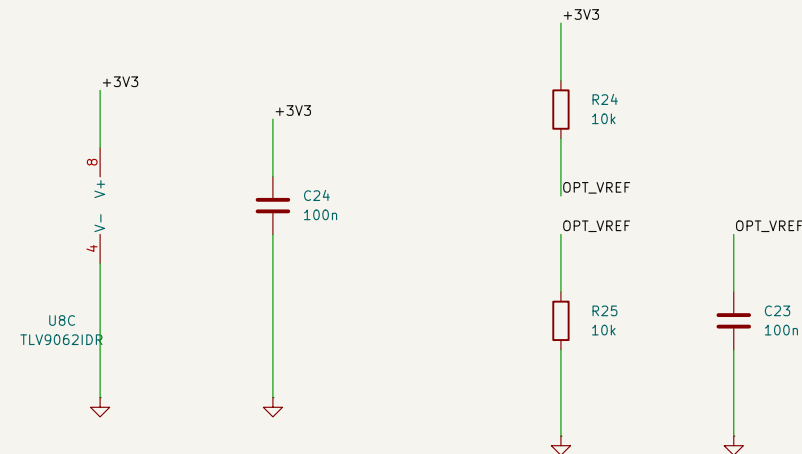
TRANSIMPEDANCE + x12 AC GAIN



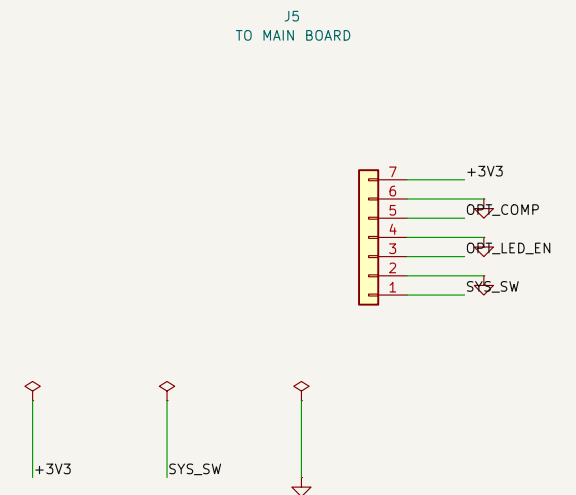
THRESHOLD COMPARATOR + HYSTERESIS



## 1.65 V REFERENCE



CABLE TO MAIN BOARD



**BalancerREF\_OptHead** – optical tach head, at 90 degrees to the main PCB on the aluminium L-bracket. One pivot bolt aims the emitter; add a second fastener (arc slot or serrated lock nut) or it drifts under vibration and loses RPM lock. The baffle between D1 and PD1 and the red acrylic window belong to this board. Designators match BalancerREF Rev C so the existing pin audits stay valid.

Cable carries DC rails and digital edges only  
Split from BalancerREF Rev C – human review required

Sheet: /  
File: BalancerREF\_OptHead.kicad\_sch

**Title: BalancerREF\_OptHead – pivoting optical tach head**

|                     |                  |         |
|---------------------|------------------|---------|
| Size: A3            | Date: 2026-09-11 | Rev: B  |
| KiCad E.D.A. 10.0.3 |                  | Id: 1/1 |